Book 2: Subject-behavioural analysis of the auditory superstitious-perception task

Auditory SP analyses

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1 What this book is and how to read it

Book 1 was about the *stimuli*: how five different acoustic-similarity metrics describe the relationships between the 300 stimuli used in the experiment, and how those relationships interact with the experimenter's target / distractor labelling. The headline result of Book 1 was that the target / distractor split explains only a tiny fraction of stimulus–stimulus variance, but it is statistically real (Fisher $F \approx 37.7$, label-axis variance fraction 0.49%, $p < 10^{-3}$).

Book 2 is about the *subjects*. Twenty-six subjects across two sites (Oberlin, $N = 16$; UChicago, $N = 10$) each saw 300 stimuli (150 in a `full_sentence` block and 150 in an `imagined_sentence` block) and gave a binary "target / not target" response per trial, for 7748 total trials.

The questions for Book 2 are:

1. **Alignment.** How well do subjects' responses agree with the experimenter's labels, separating sensitivity from response bias? (Chapter 16.)
2. **Internal consistency.** If a subject's responses are not aligned with the labels, are they nonetheless *consistent* – that is, do they treat acoustically-similar stimuli the same way? (Chapter 17.)
3. **Predictability from acoustics.** Can the five Book 1 similarity metrics predict trial-by-trial responses better than the experimenter labels alone? (Chapter 18.)
4. **Group geometry.** If we build a behavioural distance between stimuli purely from how often subjects co-assign them, does the resulting geometry look like the label geometry, or like the geometry of any acoustic metric? (Chapter 19.)
5. **Latent strategies.** Do subjects cluster into qualitatively different response strategies? (Chapter 20.)

Two pieces of context to keep in mind throughout:

- *No within-subject stimulus repeats.* A given subject sees each of their 300 stimuli exactly once, so classical test–retest reliability is not directly available. Internal consistency must therefore be defined across stimuli (chapter 17), not across repetitions.
- *Block-conditional structure.* Each of the two blocks (`full_sentence`, `imagined_sentence`) has its own independent pool of 150 stimuli. All similarity-derived features in this book are computed within block, so a stimulus is compared only to other stimuli a subject could have heard in the same block.

1.1 Trial data schema (the only column-level fact you need)

The per-subject CSVs have one row per trial with the columns:

Column	Meaning
`Subject Number`	participant ID
`Block Scheme`	e.g. `full_sentence` or `imagined_sentence`
`Stimulus Number`	integer ID of the audio file
`Stimulus Type`	**experimenter ground truth** (target / distractor)
`Subject Response`	subject's binary response (target / distractor)
`Play Count`	how many times the audio was replayed
`Trial Start Timestamp`	onset of the trial
`Play Button Clicked Timestamp`	first replay click
`Audio Start / End Timestamp`	playback boundaries
`Subject Response Timestamp`	when the response was logged

So `Stimulus Type` is the *label* (what the experimenter called it) and `Subject Response` is the *response* (what the subject called it). Throughout this book, `true_target` is the binary indicator that `Stimulus Type == ``target''` and `called_target` is the binary indicator that `Subject Response == ``target''`.

2 Chapter 16 – Signal-detection alignment and response bias

2.1 Plain-English intent

A subject can fail to match the experimenter's labels for two very different reasons. They might genuinely be unable to discriminate targets from distractors (*low sensitivity*), or they might discriminate well but prefer one response (*response bias*), e.g. calling everything "target". Signal-detection theory (SDT) gives us two numbers, d' (sensitivity) and C (criterion), that decouple these.

2.2 Intuitive formula walkthrough

Let H be the hit rate (probability of saying "target" on a real target) and F the false-alarm rate (probability of saying "target" on a distractor). Defining $z(\cdot) = \Phi^{-1}(\cdot)$ (the inverse standard normal),

$$d' = z(H) - z(F), \qquad C = -\tfrac{1}{2}\big(z(H) + z(F)\big).$$

d' measures how separated the underlying "target" and "distractor" distributions are on whatever internal axis the subject uses, in units of standard deviations. $C = 0$ means a neutral criterion; $C < 0$ is liberal (prefers "target"); $C > 0$ is conservative.

To avoid $\pm\infty$ when H or F hits 0 or 1, we use the log-linear correction (Hautus 1995): replace counts h, f over n_t, n_d trials with $\hat{H} = (h + 0.5)/(n_t + 1), \hat{F} = (f + 0.5)/(n_d + 1)$. A within-subject permutation null (label permutations) gives a per-subject p-value on d'. At the group level we test the across-subject distribution of d' against zero with a one-sided Wilcoxon signed-rank.

2.3 Result interpretation

The forest plot makes the headline result obvious: *most subjects performed at or very near chance.* Across all 52 (subject $\times$ block) cells the mean accuracy is 0.510 (range 0.42–0.58), the mean hit rate is 0.480 and the mean false-alarm rate is 0.460.

At the group level there is still a small but reliable signal in the `full_sentence` block: mean $d' = 0.091$, Wilcoxon one-sided $p = 0.006$, t-test against zero $t = 2.61$, $p = 0.007$. The `imagined_sentence`

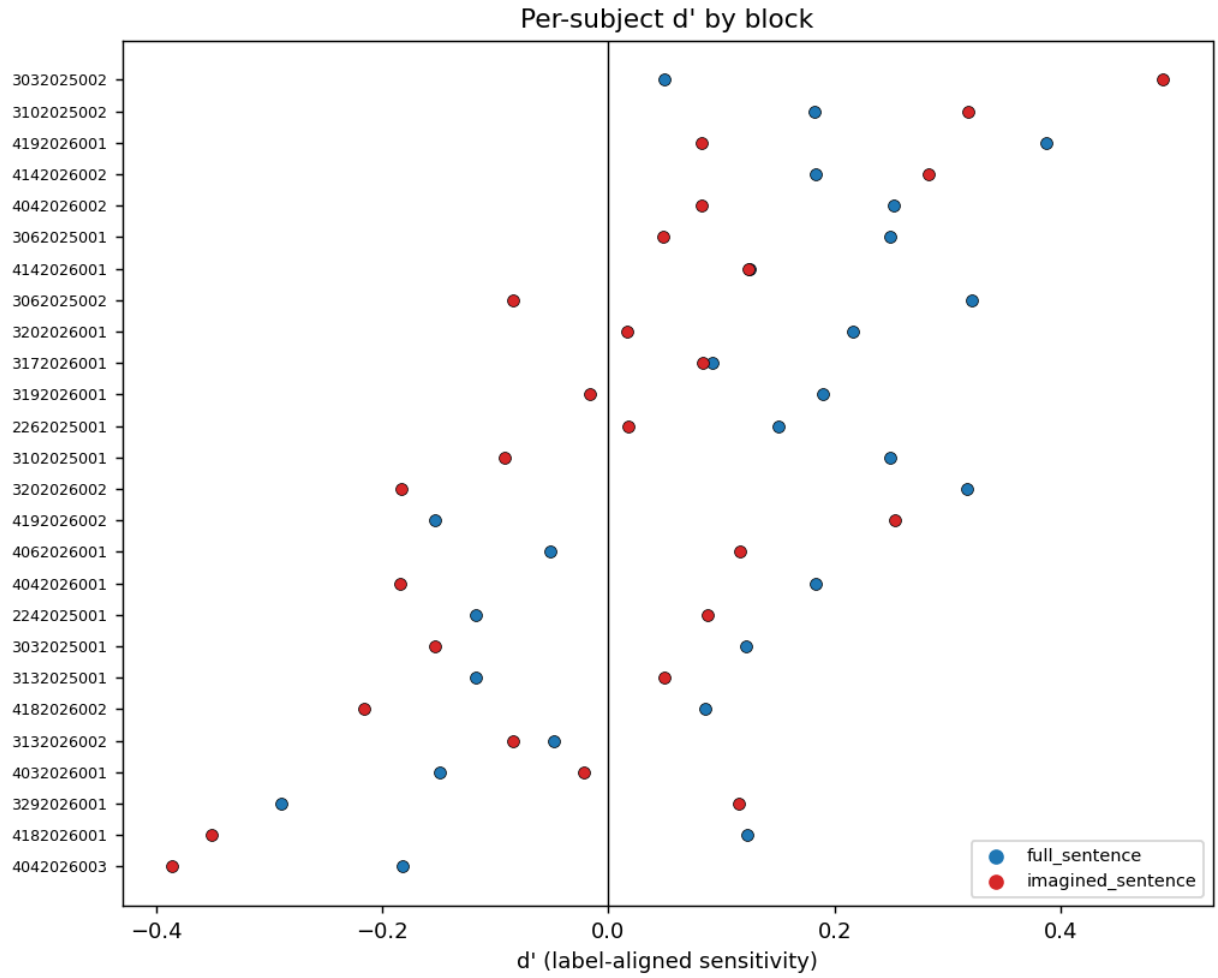

Figure 1: Per-subject d' in each block. Each row is a subject; blue is `full_sentence`, red is `imagined_sentence`. Most d' values cluster tightly around zero.

block has mean $d' = 0.015$ and is statistically indistinguishable from chance ($p = 0.38$). Only 1/26 subjects in the imagined block had a permutation-significant d'; none did in the full block individually. The group-level d' in the full block is therefore real but distributed across all subjects rather than concentrated in a few strong performers.

2.4 Reader takeaway

Subjects as a group do barely above chance on the `full_sentence` block, and at chance on the `imagined_sentence` block. There is a small conservative bias in the imagined block. Any further analysis must therefore explain a behavioural signal that is genuinely small.

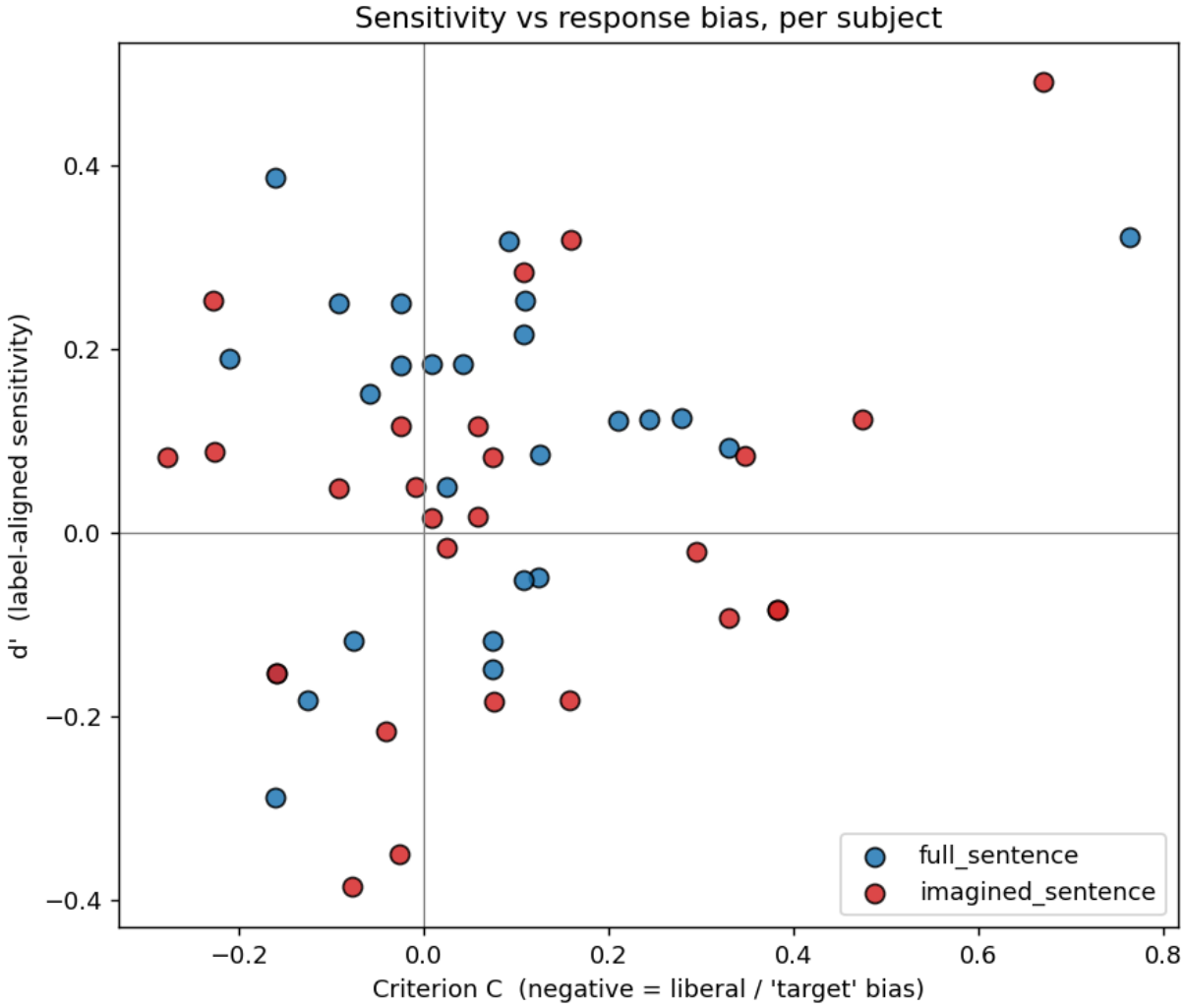

Figure 2: Sensitivity (d') versus response bias (C). Slight conservative bias in the imagined block (mean $C = 0.094$, $t = 2.06$, $p = 0.05$); near-neutral in the full block.

3 Chapter 17 – Internal response consistency

3.1 Plain-English intent

If a subject is at 51% accuracy, they could either be (a) randomly guessing or (b) using a stable but "wrong" rule. We cannot distinguish these from accuracy alone, but we *can* ask whether stimuli that are acoustically near each other (under some metric) tend to receive the same response from that subject. If yes, the subject is responding consistently to acoustic structure even when not matching the labels.

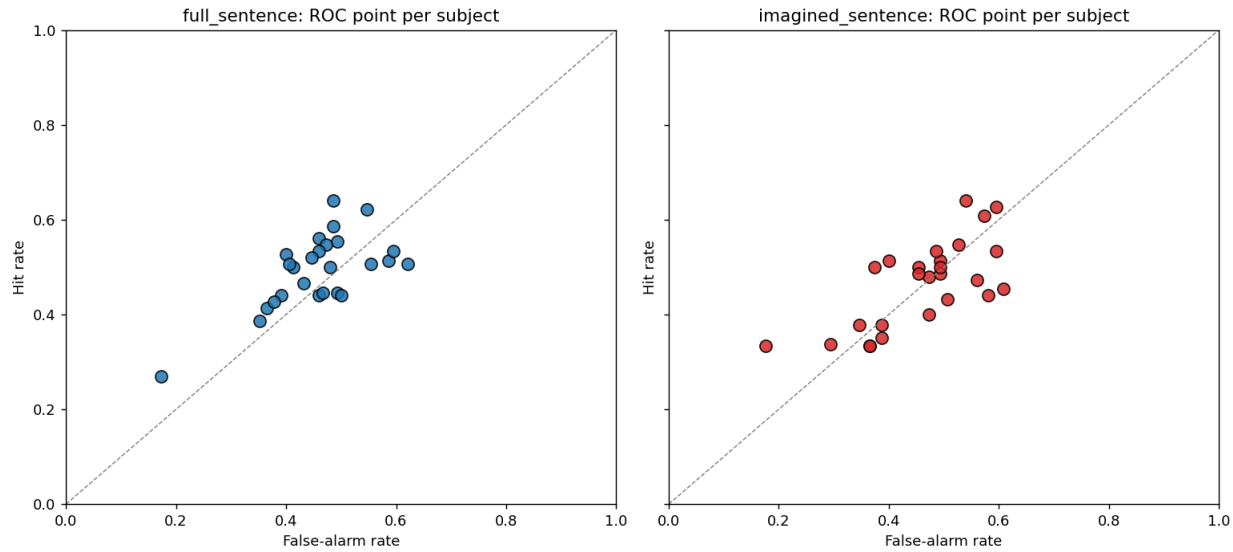

Figure 3: Each subject's (FA, hit) point, by block. Points hug the diagonal, again indicating near-chance discrimination.

3.2 Intuitive formula walkthrough

For each metric m and each block, we build a k-nearest-neighbour graph on the 150 stimuli. Then for each subject s we compute

$$\mathrm{cons}_{s,m} = \frac{1}{N\,k} \sum_{i=1}^{N} \sum_{j \in \mathrm{kNN}(i)} \mathbf{1}\big[Y_{s,i} = Y_{s,j}\big],$$

the average rate at which two acoustically near-neighbour stimuli get the same response from subject s. A within-subject permutation null is built by shuffling that subject's response vector over the 150 stimuli, which preserves the marginal response rate. We use $k = 10$, 500 permutations.

3.3 Result interpretation

Across both blocks and all five metrics, the per-subject neighbour consistency clusters tightly around 0.51 (range of group means 0.506–0.519 across the ten metric–block cells). This is essentially indistinguishable from the chance value of 0.50 that would be expected if neighbour structure carried no information about the subject's response choices.

3.4 Reader takeaway

There is no evidence that subjects are using any acoustic-similarity neighbourhood – under any of the five metrics from Book 1 – as the basis for their responses. Combined with chapter 16, this rules out the "consistent but unlabelled rule" explanation: subjects are not just using a different stable rule based on these acoustic features, they are giving near-random responses with respect to acoustic neighbourhood.

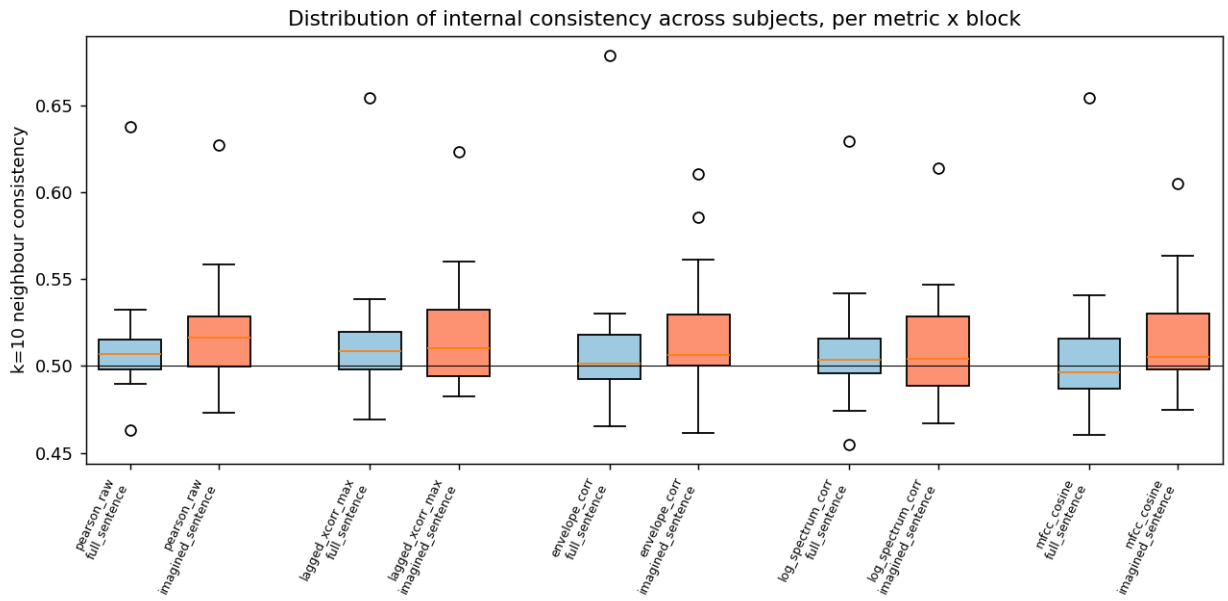

Figure 4: Distribution across subjects of $k = 10$ neighbour consistency, per metric and block. All medians sit near the chance value of 0.5.

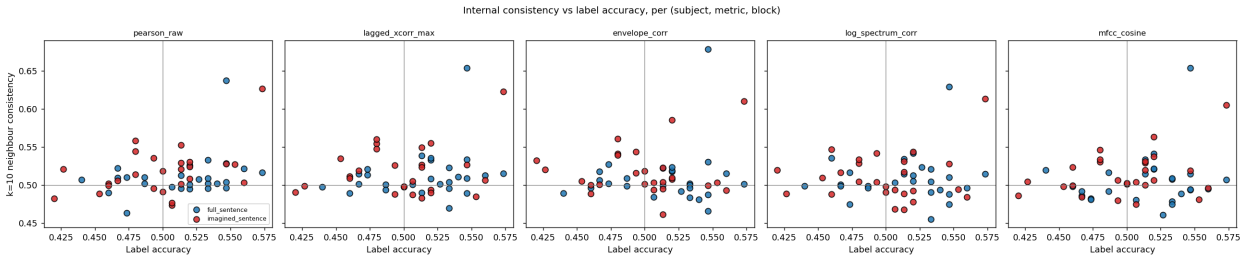

Figure 5: Per-subject internal neighbour consistency vs. label accuracy, faceted by metric. Quadrants from the planning report: top-right (consistent and accurate), top-left (consistent but mis-labelled rule), bottom-right (accurate but inconsistent), bottom-left (chance). Almost every subject lives near the bottom-left intersection.

4 Chapter 18 – Trial-level logistic prediction

4.1 Plain-English intent

The neighbour-consistency analysis is a coarse k-NN summary. A logistic regression is the natural smooth, multi-feature version of the same question: given each stimulus's relationship to the rest of its block (under each acoustic metric) and the trial number, can we predict whether this subject will say "target"?

4.2 Intuitive formula walkthrough

For each metric m, define the within-block target–minus–distractor contrast for stimulus i:

$$x_i^{(m)} = \frac{1}{|T_b \setminus \{i\}|} \sum_{j \in T_b \setminus \{i\}} S_{ij}^{(m)} - \frac{1}{|D_b \setminus \{i\}|} \sum_{j \in D_b \setminus \{i\}} S_{ij}^{(m)},$$

where T_b, D_b are the target / distractor sets of stimulus i's block and $S^{(m)}$ is the metric-m similarity matrix. Positive $x_i^{(m)}$ means stimulus i is more similar to the block's targets than to its distractors under metric m.

For each (subject, block) we fit

$$P(Y_t = 1 \mid x_t, \tau_t) = \sigma\big(\beta_0 + \sum_m \beta_m \, x_t^{(m)} + \beta_\tau \tau_t\big),$$

where τ_t is the within-block trial index z-scored. We use 5-fold stratified CV and report the held-out AUC.

4.3 Result interpretation

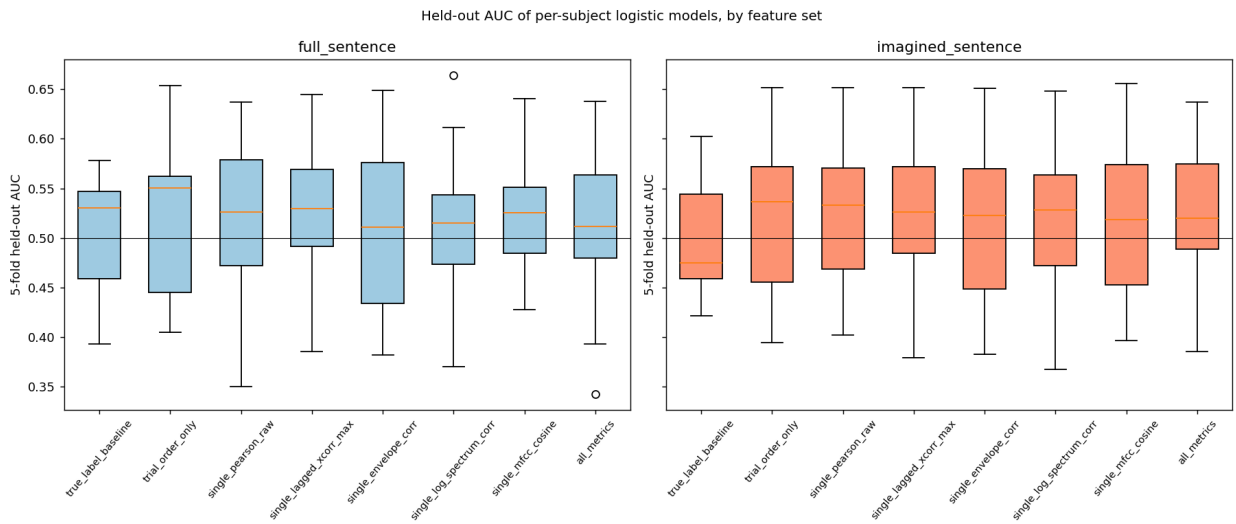

Figure 6: Per-subject 5-fold held-out AUC for several feature sets, by block. The $y = 0.5$ line is chance.

The mean held-out AUCs sit in the very narrow range 0.50–0.53. Crucially, the `trial_order_only` model – which uses the within- block trial index alone – already achieves ~ 0.52 AUC on average, matching or beating every single-metric model. The `true_label_baseline` model, which uses only the experimenter's ground-truth label, gives 0.505 in the full block and 0.498 in the imagined block. The combined `all_metrics` model does not help either (0.509 full, 0.520 imagined).

4.4 Reader takeaway

The five Book 1 acoustic-similarity metrics offer essentially no predictive purchase on individual responses. The fact that a one-feature `trial_order` model is competitive suggests that what little structure exists in the responses is closer to a slow drift in liberal/conservative tendency over a block than a stimulus-locked acoustic-feature dependence.

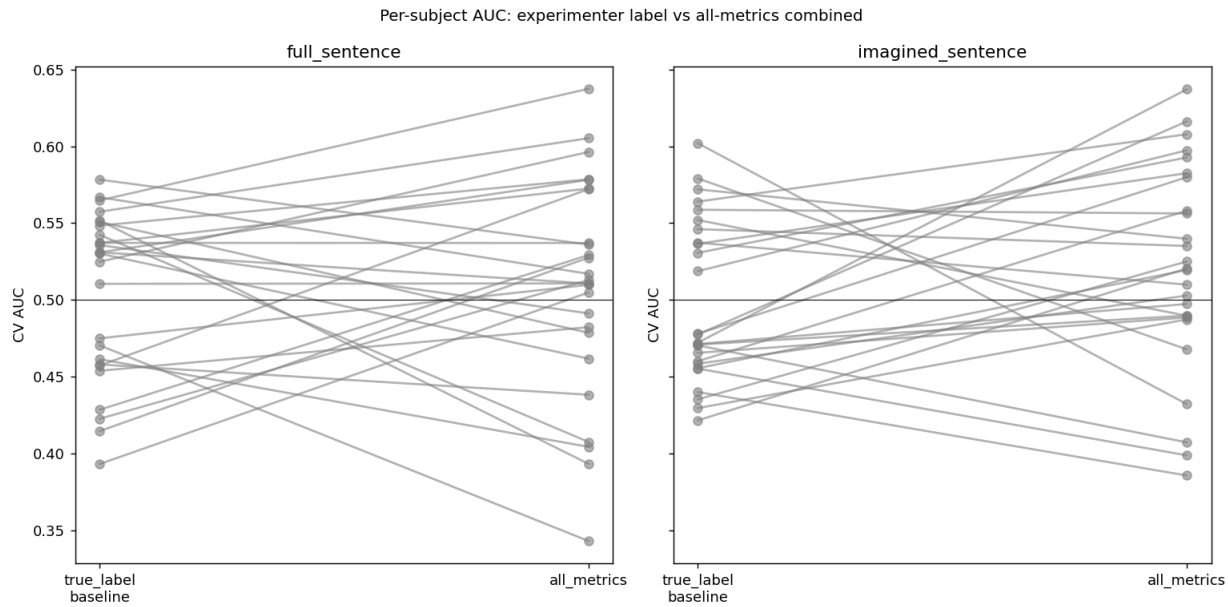

Figure 7: Paired per-subject AUC: the experimenter true label (left) vs. the all-metrics combined model (right). The all-metrics model is at best a hair better.

5 Chapter 19 – Response geometry and RSA

5.1 Plain-English intent

Even if individual subjects are noisy, perhaps *across* subjects there is a stable answer to "which pairs of stimuli get the same response?". We build that answer as a behavioural co-assignment matrix and ask which of our several stimulus geometries (label, Pearson, lagged xcorr, envelope, log spectrum, MFCC) it most resembles.

5.2 Intuitive formula walkthrough

For each block and each pair (i, j) of stimuli,

$$C_{ij} = \frac{1}{|S_{ij}|} \sum_{s \in S_{ij}} \mathbf{1}\big[Y_{s,i} = Y_{s,j}\big],$$

where S_{ij} is the set of subjects who responded on both stimuli. We treat $1 - C$ as a behavioural representational dissimilarity matrix (RDM), then:

- Cluster on $1 - C$ (Agglomerative, average linkage, $k \in \{2, 3, 4, 5\}$) and report ARI / NMI vs. the experimenter labels.
- RSA: Spearman correlation of the upper triangle of $1 - C$ with the upper triangle of each model RDM. Permutation p values come from shuffling stimulus indices on the model side (2000 permutations).

5.3 Result interpretation

The Adjusted Rand Index of behaviour-derived clusters against the experimenter labels is ≤ 0.01 in absolute value for every k in both blocks. RSA correlations against all six model RDMs (label plus

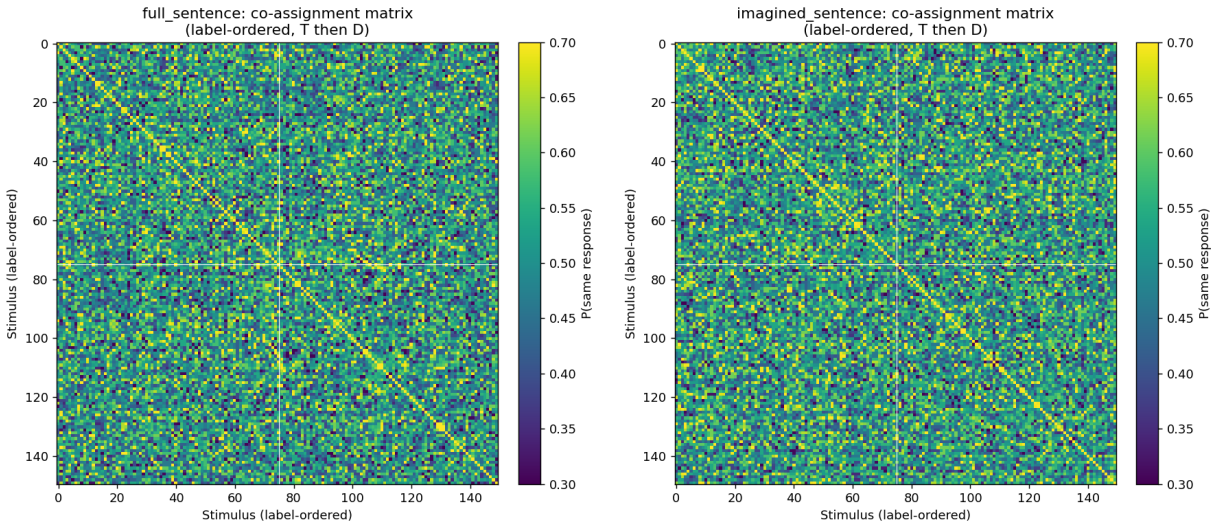

Figure 8: Behavioural co-assignment matrices, with stimuli ordered so that all targets precede all distractors (white lines mark the boundary). If behaviour tracked the labels, the diagonal blocks would be visibly lighter than the off-diagonals; they are not.

five acoustic metrics) cluster around $|r| < 0.005$ and are not significant under permutation, with one curious exception: in the `imagined_sentence` block, the behavioural RDM correlates *negatively* with the log-spectrum RDM ($r = -0.029$, $p = 0.006$), i.e. stimuli with similar log spectra are slightly *less* likely to share responses. The effect is tiny and we flag it only as a hypothesis to revisit if more data become available.

5.4 Reader takeaway

The geometry of subject responses is essentially flat with respect to all six candidate stimulus geometries. Whatever drives the per-subject responses is not shared at the group level in any way that respects acoustic neighbourhood or experimenter labelling.

6 Chapter 20 – Latent response strategies

6.1 Plain-English intent

Maybe subjects *do* use distinct strategies, just at the individual level. If so, clustering subjects by their full response vector (across all 150 stimuli of a block) might reveal e.g. a "yes-saying" group and a "label-tracking" group. We complement that with clustering on the metric-weight profiles fit per subject in chapter 18, in case different subjects rely on different acoustic cues even when their overall accuracy is the same.

6.2 Intuitive formula walkthrough

(A) Response-vector clustering. Build the subject $\times$ stimulus matrix Y, mean-centre rows, compute pairwise cosine similarity, rescale to $[0, 1]$ as an affinity, and run spectral clustering for $k \in \{2, 3, 4\}$.

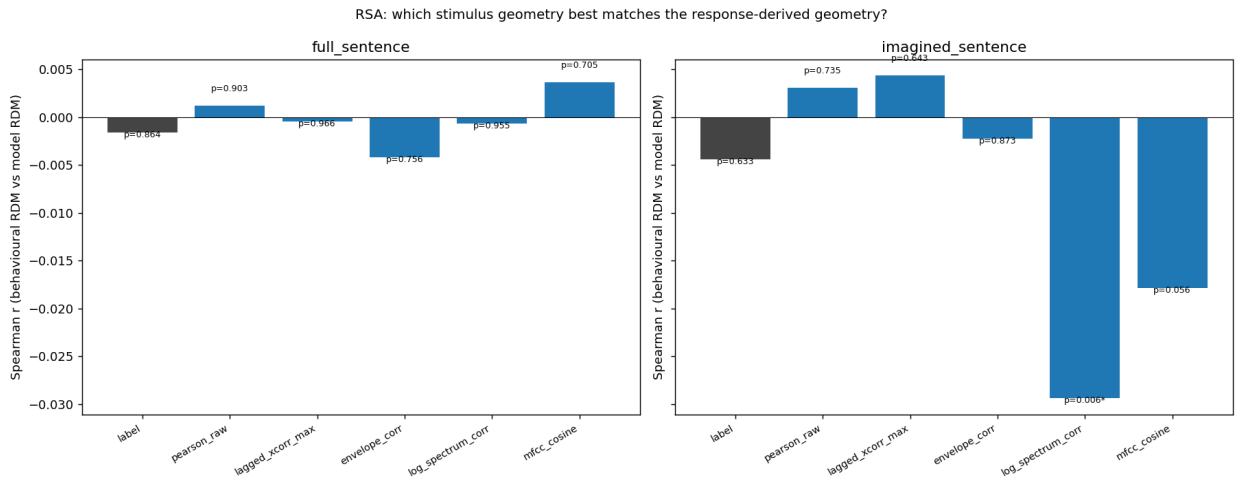

Figure 9: RSA Spearman correlations between the behavioural RDM and each model RDM, by block, with permutation p-values annotated.

(B) Coefficient-profile clustering. For each subject fit a five-feature logistic regression (one feature per metric, standardised) on the subject's trials. Z-score the resulting weight vectors across subjects and run K-means with $k \in \{2, 3, 4\}$.

6.3 Result interpretation

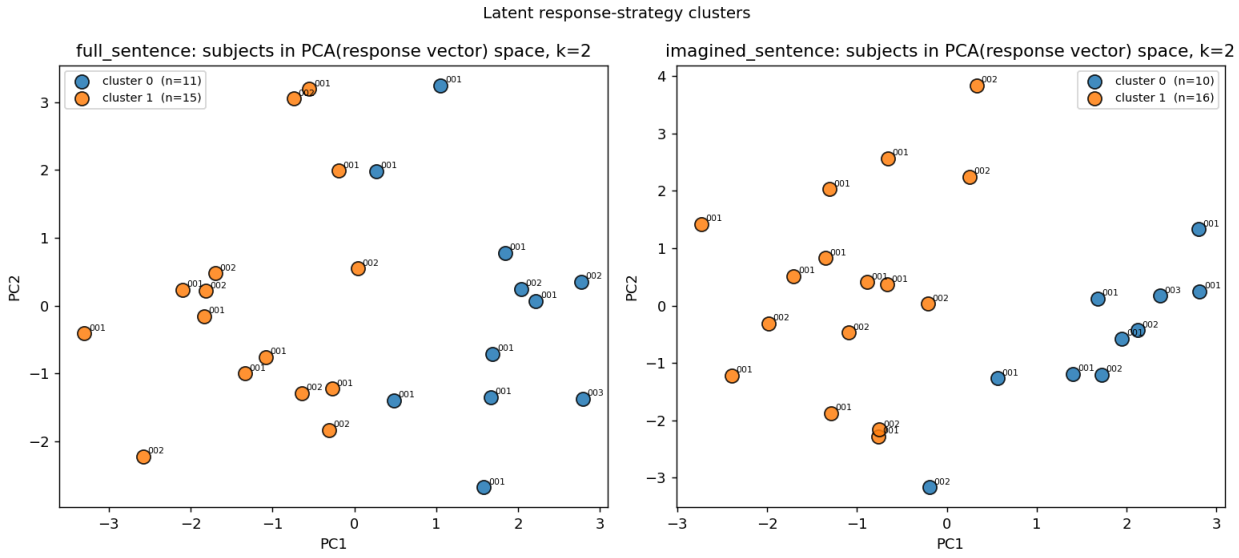

Figure 10: Subjects in the first two principal components of their response vector, coloured by spectral cluster ($k = 2$). The clusters reflect mostly a single axis of slightly liberal vs. slightly conservative response rate.

The $k = 2$ response clusters in the `full_sentence` block split the 26 subjects into groups of 11 and 15 subjects with mean response rates 0.493 and 0.463, mean accuracies 0.522 and 0.508. In the `imagined_sentence` block the groups are 10 vs. 16 subjects with mean response rates 0.466

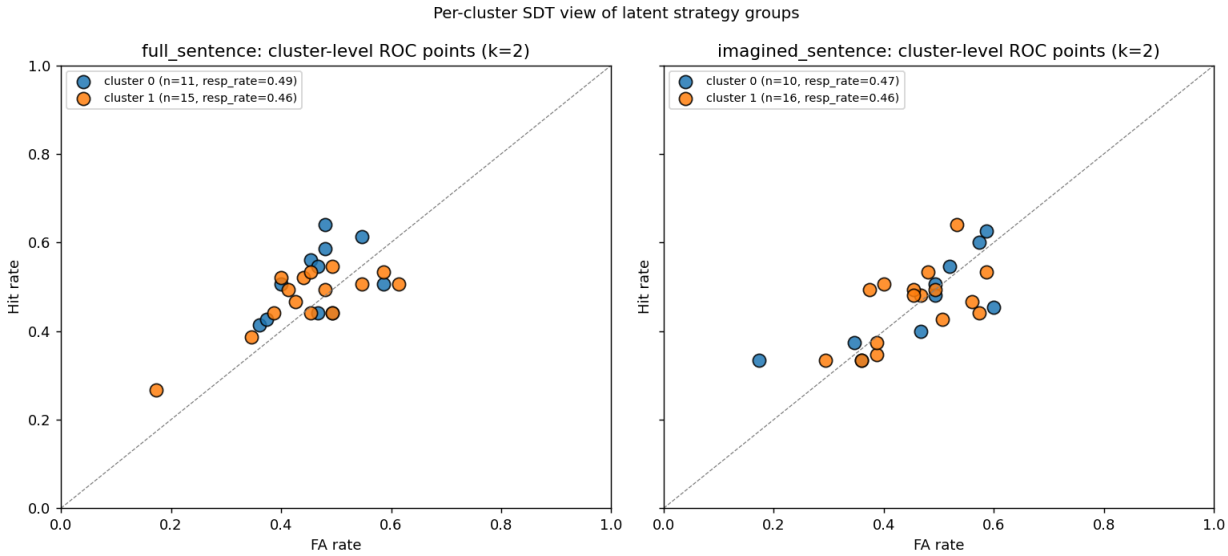

Figure 11: ROC points by subject, coloured by the $k = 2$ response-vector cluster. Cluster centroids are very close together; clusters do not correspond to meaningfully distinct sensitivities.

vs. 0.462 and mean accuracies 0.498 vs. 0.500. Translation: the clusters reflect a tiny amount of structure in subjects' overall response rate but no qualitative strategy difference.

6.4 Reader takeaway

There is no evidence for distinct, acoustically-defined response strategies across these 26 subjects. The mild bimodality the clusterer finds is explained almost entirely by small differences in overall response rate.

7 Synthesis

The five chapters of Book 2 paint a coherent and empirically tight picture:

- **Subjects are at chance overall**, with a small but real group-level d' of 0.09 in `full_sentence` ($p = 0.006$) and $d' \approx 0$ in `imagined_sentence`.
- **Subjects' responses are not consistent with respect to any Book 1 acoustic-similarity neighbourhood** – the $k = 10$ neighbour consistency sits at 0.51 for every metric in every block, where 0.50 is chance.
- **Logistic models built from the Book 1 metrics do not predict subject responses better than chance** (held-out AUC ≤ 0.53 in all cases, including a model that combines all five metrics; a model that uses only the within-block trial index does just as well).
- **The behavioural co-assignment geometry is essentially flat** against all six candidate model geometries (Spearman $|r| < 0.03$, mostly < 0.005; ARI vs. labels $| < 0.01|$).
- **No distinct response strategies cluster the subjects** – attempted clusterings differ only in overall response rate, with no meaningful sensitivity gap.

A natural reading of these five results together is: *the experiment as designed produces near-random responses*, and the small label-related signal in the `full_sentence` block is distributed broadly across subjects rather than carried by a competent subgroup. Because the Book 1 metrics already showed that the label axis explains less than 0.5% of stimulus-pair variance (chapter 8), this is consistent with the psychophysical task being above human discrimination threshold for these stimuli rather than a failure of any single subject or analysis.

Limitations and what would change the picture

- *No within-subject repetitions.* Direct test–retest reliability is unavailable, so all consistency claims are at the across-stimulus level (chapter 17) or across-subject (chapter 19).
- *Modest sample.* 26 subjects $\times$ 150 trials per block gives the group test reasonable power for moderate effects (the $d' = 0.09$ effect is detectable), but small per-subject AUC differences of ~ 0.02 are at the edge of detectability with 5-fold CV on 150 trials.
- *Acoustic feature space.* We tested five metrics from Book 1. A negative result against *these* five does not preclude richer learned representations (e.g. self-supervised speech embeddings) revealing structure, and is a natural next analysis if the question is pursued further.

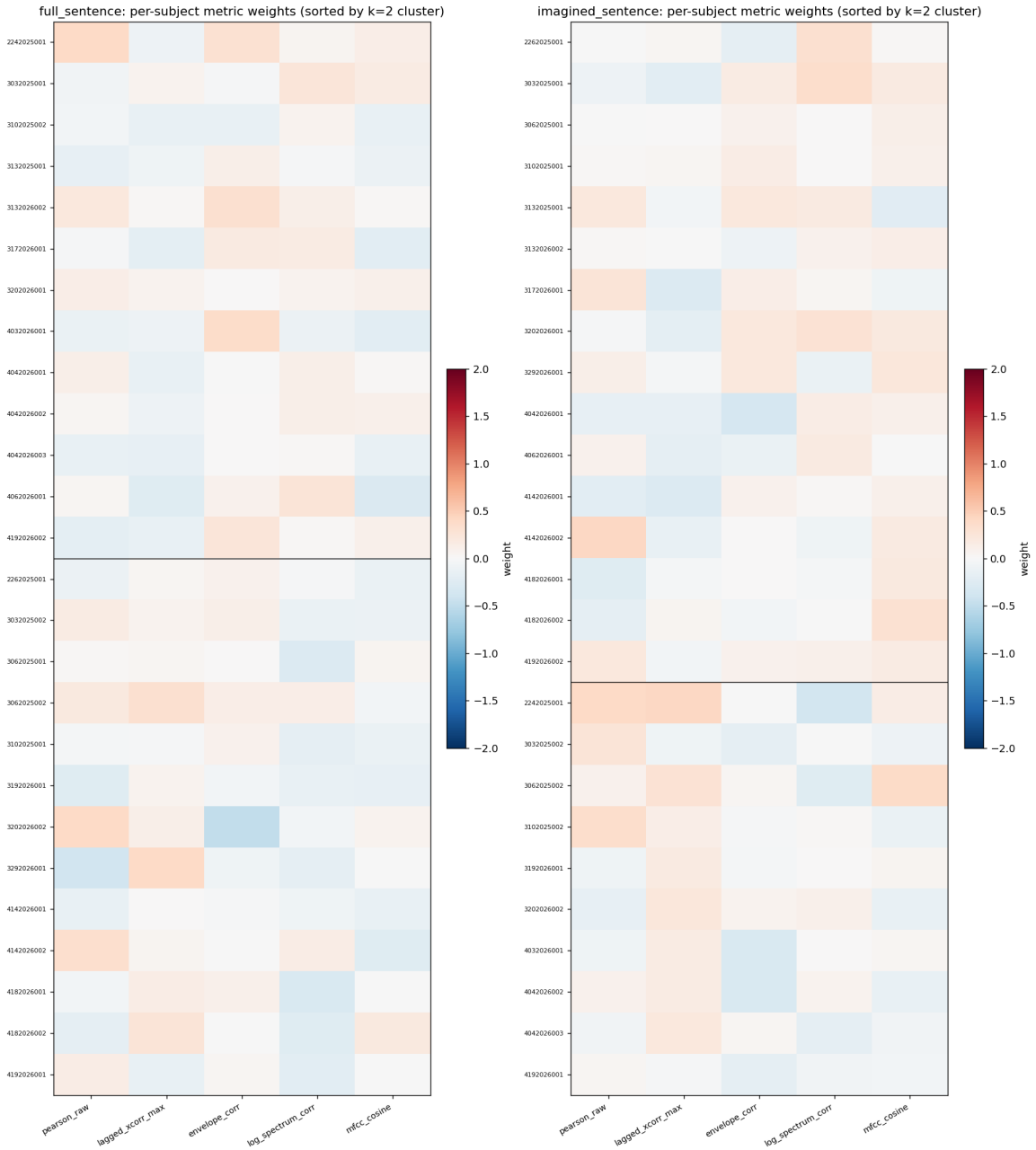

Figure 12: Per-subject metric weights from the logistic regressions (rows are subjects, columns are metrics), with rows sorted by the $k = 2$ cluster on the weights. The visible vertical structure is not a coherent "strategy" but rather noise around zero – consistent with the chapter 18 finding that all metric features are near-uninformative.